

Model Output

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Travel models generate large amounts of output. This model focuses on the organization of that output to make the information more accessible. First, all outputs are stored in the "outputs" folder inside the scenario directory. Input files are not modified during a model. Second, output table fields include a field description. Hover over the field to see the following info:

- Which model step the field came from
- What information the field contains
- What parameter file created the field

In general, model outputs are organized as follows:

- Network data is stored on the output network
 - At the end of the model, a loaded network is created in the summary folder
- Zonal calculations (area type, productions, attractions, etc.) are stored in the output se data file
 - Field descriptions provide valuable information about each field
- Trip matrices are stored in folders by model step (distribution, mode, assignment, etc.)

The sections below go over the most important output folders in more detail.

Summary Folder

This folder is where every scenario review should start. It contains all the summarized tables, maps, and other data that is created with the intent of conveying model results to the user.

loaded_network

This folder contains the output network with all assignment results appended. In addition, further calculations have been done to create daily fields for various metrics like delay that are not standard assignment output. This file is used to generate many of the summary maps and reports, but can also be useful for custom summaries the user might want to perform.

maps

This folder contains several maps that fall into two categories:

- Count comparison maps
 - Helpful during calibration/validation. Not meaningful in future years.
- V/C Maps
 - These provide volume and congestion numbers for each time period as well as daily. In the daily map, the highest V/C during the day is shown.

Assignment_Validation.html

This report compares model results to traffic counts, and is only meaningful in the base year when traffic counts were collected. It provides a wealth of comparison metrics and plots to measure the assignment performance.

link_summary_by_FT_and_AT.csv

Often, regional metrics are compared between scenarios (e.g. how does total delay change from one scenario to another). This file facilitates these comparisons by providing a summary of these common metrics by scenario. They are segmented by facility type and area type, but can be quickly summed into totals as well.

Other Summary Folders

The other subfolders in the summary directory are generated primarily to support rmarkdown pages and R summaries. For example, R cannot read .mtx files, so they are exported to CSVs in long format. Not only did this facilitate using rmarkdown for model development documentation, but it also supports hosting future application projects in a similar fashion.

Networks Folder

The first steps of the model process the network file to calculate speeds and capacities. The results are stored in fields on this network.

SE data Folder

Results from trip generation and time of day are stored in this file (although some intermediate generation files are also stored in the "generation" output folder).

Assignment Folder

Because the assignment process is more complex than most models, the assignment folder contains some additional information in case questions arise that cannot be answered using

contains some additional information in case questions arise that cannot be answered using the reports in the summary folder.

RMSE Reports

An RMSE report is generated for each time period that tracks the progress of assignment. For each cycle, the RMSE and %RMSE are shown. These can be consulted to determine how tightly each period converged.

Assignment Matrices

The final demand matrices fed into assignment are stored in this folder to make it clear what was run. Each core in the matrix is an assignment class.

Cycle Folders

In addition to the RMSE reports, the output of each assignment cycle is stored. These provide the ultimate level of detail. Importantly, they also contain the path files created during assignment. This supports easy select link analysis without having to re-run assignment. Go to Planning -> Utilities -> OUE Select Link Analysis to learn more about how to use path files to create select link plots.

Finally, remember that the model performs feedback cycles independently for each time period. That means that some of the later cycles may not contain every period. This just means that those periods converged in fewer cycles.

Skims Folder

This model tracks both link volume as well as skim time across cycles. Unlike assignment, only the previous cycle's skim matrices are maintained. This was done primarily to save hard drive space given the limited utility of having every interim skim matrix.

Other Output Folders

The remaining files are named and organized into folders to make them easy to find and understand. For example, all distribution output can be found in the distribution folder. The files are named in such a manner that they are organized when sorted by name (e.g. they start with "utilities", "probabilities", or "trips"). They are reviewed much less frequently - generally only for very specific applications or during model development.

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User's Guide



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